

# Measurement & Evaluation: Bloom, Reliability, Validity, Statistics

मापन एवं मूल्यांकन: ब्लूम, विश्वसनीयता, वैधता, सांख्यिकी

## Measurement & Evaluation: Bloom, Reliability, Validity, Statistics

**What UPSC asks on this** - Terms and who coined them: measurement vs assessment vs evaluation; formative/summative (Scriven); norm- vs criterion-referenced (Glaser); assessment *of*, *for* and *as* learning. - Bloom's taxonomy: the order of levels, original 1956 vs revised 2001, the three domains and their authors. - Reliability and validity: the types, what each method measures, the Spearman-Brown and SEM formulas, and how the two relate. - Item analysis: difficulty index and discrimination index, with cut-offs. - Statistics: mean/median/mode, SD, normal-curve percentages, z and T scores, percentiles, skewness, Spearman's rho. - Policy: CCE under RTE, NEP 2020 paras 4.34–4.41, PARAKH, the Holistic Progress Card, PARAKH Rashtriya Sarvekshan 2024, two board exams.

## 1. Measurement, assessment, evaluation

**Measurement gives a number; evaluation adds a judgement of worth against objectives.**

- **Measurement** assigns numbers to a trait by a rule (62 out of 100). It is purely quantitative and says nothing about "good" or "bad".
- **Assessment** is the wider process of collecting evidence of learning: tests, observation, portfolios, projects. The word comes from Latin *assidere*, "to sit beside".
- **Evaluation** = measurement + value judgement. "62 means she has met the learning outcome on fractions." It uses both quantitative and qualitative evidence.

**Ralph Tyler**, called the father of educational evaluation, tied objectives, learning experiences and evaluation into one cycle. Evaluation exists to tell you whether the objectives were achieved, so objectives must be written first.

Delhi example: a Class 8 mid-term score is measurement. The teacher's remedial grouping based on it is assessment for learning. The end-of-year promotion decision is evaluation.

## Kinds of evaluation by purpose

| Type       | When                    | Purpose                                   |
|------------|-------------------------|-------------------------------------------|
| Placement  | Before teaching         | Entry behaviour, grouping                 |
| Formative  | During teaching         | Feedback to improve teaching and learning |
| Diagnostic | When errors persist     | Locate the exact weakness and its cause   |
| Summative  | End of unit, term, year | Grade, certify, promote                   |

- **Michael Scriven (1967)** coined *formative* and *summative*. **Robert Glaser (1963)** coined *norm-referenced* and *criterion-referenced*.
- **Prognostic** evaluation predicts future success; an aptitude test is prognostic.
- **Assessment of learning** = summative. **Assessment for learning** = formative, feedback to teacher and learner. **Assessment as learning** = the learner monitors their own learning through self- and peer-assessment (metacognition). NEP 2020 shifts weight to the last two.

Memory aid: when the cook tastes the soup it is formative; when the guest tastes it, summative.

## Norm-referenced vs criterion-referenced

- **Norm-referenced (NRT)**: a score gets meaning from the group: percentile rank, class rank, grades awarded by position. It needs a spread of scores, so items of medium difficulty are preferred. Used for selection.

- **Criterion-referenced (CRT):** a score gets meaning from a fixed standard: "reads 60 words a minute", "attained Learning Outcome M-5.3". No comparison with peers. Used in mastery learning, NIPUN Bharat targets and driving tests. Items may all be easy if the standard is basic.

Trap: "top 10% of the class" is norm-referenced; "8 of 10 sums correct" is criterion-referenced. A fixed pass mark of 33% is criterion-referenced; a grade given by rank in the group (positional grading) is norm-referenced.

## **2. Bloom's taxonomy and the three domains**

**Bloom gives the ladder of objectives; the 2001 revision turned nouns into verbs and put Create on top.**

Benjamin Bloom and colleagues published *Taxonomy of Educational Objectives, Handbook I: Cognitive Domain* in **1956**. Six levels, each resting on the one below.

| Level          | Original 1956<br>(nouns) | Revised 2001<br>(verbs) | Sample verbs<br>(revised)              |
|----------------|--------------------------|-------------------------|----------------------------------------|
| 1<br>(lowest)  | Knowledge                | Remember                | define, list,<br>recall                |
| 2              | Comprehension            | Understand              | explain,<br>summarise,<br>classify     |
| 3              | Application              | Apply                   | solve, use,<br>demonstrate             |
| 4              | Analysis                 | Analyse                 | compare,<br>differentiate,<br>organise |
| 5              | Synthesis                | Evaluate                | judge, critique,<br>justify            |
| 6<br>(highest) | Evaluation               | Create                  | design,<br>construct,<br>compose       |

Read rows 5 and 6 as positions on the ladder, not renamings: Synthesis was renamed **Create** and moved up to level 6, and Evaluation became **Evaluate** at level 5.

The revision by **Lorin Anderson and David Krathwohl (2001)** made three changes. Nouns became verbs. *Synthesis* became **Create** and moved to the top, above **Evaluate**. A second axis was added, the **knowledge dimension**: Factual, Conceptual, Procedural, **Metacognitive**. Every objective is now a cell such as "Apply procedural knowledge".

Remember, Understand and Apply are **lower-order thinking skills (LOTS)**. Analyse, Evaluate and Create are **higher-order thinking skills (HOTS)**, the target of CBSE's competency-based questions.

```
`mermaid flowchart TD
  A["Remember: recall facts"] --> B["Understand: explain ideas"]
  B --> C["Apply: use in new situation"]
  C --> D["Analyse: break into parts"]
  D --> E["Evaluate: judge and justify"]
  E --> F["Create: produce new work"]`
```

The other two domains:

- **Affective domain** (Krathwohl, Bloom and Masia, 1964, Handbook II): Receiving → Responding → Valuing → Organising → Characterising by a value complex. Mnemonic: **R-R-V-O-C**.
- **Psychomotor domain**: Bloom's group never finished it. **Simpson (1972)**: Perception, Set, Guided response, Mechanism, Complex overt response, Adaptation, Origination. **Dave (1970)**: Imitation, Manipulation, Precision, Articulation, Naturalisation (mnemonic **IMPAN**). **Harrow (1972)** is the third version.

Delhi example: the Science objective "students will design a low-cost water filter" is *Create* (cognitive), *Precision* (psychomotor) and *Valuing* (affective, caring about water).

Memory aid for the revised order: **R-U-A-A-E-C**, "Rats Usually Avoid Angry Energetic Cats".

### 3. What makes a good test

**Validity** asks "does it measure what it claims?";

**reliability** asks "does it measure consistently?"

Qualities of a good test: **validity, reliability, objectivity, usability** (time, cost, ease of scoring), **norms, discriminating power, comprehensiveness.**

Objectivity means the score does not depend on who marks it.

#### Validity

| Type                   | Question it answers                                  | How established                                       |
|------------------------|------------------------------------------------------|-------------------------------------------------------|
| Content                | Does it cover syllabus and objectives in proportion? | Blueprint; expert judgement                           |
| Face                   | Does it look like it measures the trait?             | Surface inspection (weakest)                          |
| Concurrent (criterion) | Does it agree with another measure taken now?        | Correlate with an existing test                       |
| Predictive (criterion) | Does it forecast later performance?                  | Correlate with a later criterion                      |
| Construct              | Does it behave as the theory of the trait predicts?  | Convergent and discriminant evidence, factor analysis |

Validity is reported as a **validity coefficient**, a correlation with the criterion. Validity drops with unclear directions, hard vocabulary, ambiguous items, wrong difficulty level, too few items, answer patterns

and a mismatch between what was taught and what is tested.

**Reliability**

Reliability is consistency of scores, reported as a **reliability coefficient** between 0 and 1. A classroom test should reach about **0.70**; a standardised test about **0.90**.

| Method                        | Measures                                 | Note                                        |
|-------------------------------|------------------------------------------|---------------------------------------------|
| Test-retest                   | Stability over time                      | Memory and practice effects                 |
| Parallel (equivalent) forms   | Equivalence of two forms                 | Hardest to build, best method               |
| Split-half                    | Internal consistency, one sitting        | Odd-even split; correct with Spearman-Brown |
| Kuder-Richardson KR-20, KR-21 | Internal consistency, right/wrong items  | KR-21 assumes equal item difficulty         |
| Cronbach's alpha (1951)       | Internal consistency, rating-scale items | Generalises KR-20                           |
| Inter-rater                   | Agreement between scorers                | Essays, rubrics                             |

Reliability rises with **more items**, **moderate difficulty**, a **heterogeneous group** (wider spread of true scores), **objective scoring**, clear items and good conditions. Speeded tests inflate split-half reliability.

**Spearman-Brown prophecy formula:**  $r_n = n \cdot r \div [1 + (n - 1) \cdot r]$ , where  $n$  is the factor by which the test is lengthened. For split-half,  $n = 2$ :  $r_{full} = 2r \div (1 + r)$ .

**Standard error of measurement:**  $SEM = SD \times \sqrt{1 - r}$ . It turns a reliability figure into score units: the band in which a student's true score probably lies.

### How the two relate

- Reliability is **necessary but not sufficient** for validity. A reliable test can be invalid (a weighing scale always 2 kg heavy is consistent but wrong). An unreliable test can never be valid.
- **Maximum validity =  $\sqrt{\text{reliability}}$**  (the index of reliability). Reliability 0.64 caps validity at 0.80.
- Objectivity supports reliability, which supports validity.

Memory aid, the dartboard: darts clustered away from the bull are reliable but not valid; scattered around the bull, neither; clustered on the bull, both.

## 4. Building and refining a test

**A test grows from objectives through a blueprint, then is tried out and refined by item analysis.**

```
`mermaid flowchart TD
  A["Define objectives"] --> B["Prepare blueprint"]
  B --> C["Write items"]
  C --> D["Try out on sample"]
  D --> E["Item analysis: p and D"]
  E --> F["Check reliability and validity"]
  F --> G["Prepare norms"]`
```

- **Blueprint (table of specifications):** a grid of objectives (Bloom levels)  $\times$  content units  $\times$  question

types × marks. It is the main guarantee of **content validity**. CBSE's Assessment and Evaluation handbook builds every paper from one.

- **Item types:** selection type (MCQ, true/false, matching) and supply type (completion, short answer, essay). An MCQ has a **stem**, a **key** and **distractors**. Objective items score reliably but take long to write; essays are the reverse.
- **Standardised test:** fixed content, administration, scoring and norms, produced through try-out and item analysis. A **teacher-made test** serves one classroom and usually has no norms.
- **Other tools:** rubrics (**analytic** = criterion-wise scores; **holistic** = one overall score), portfolios, anecdotal records, checklists, rating scales. The Holistic Progress Card relies on these.

### Item analysis

**Difficulty index** (facility value)  $p = \frac{\text{number correct}}{\text{number attempting}}$ , often as a percentage. Higher p means an easier item. For norm-referenced tests the usual target is **0.30 to 0.70**, with 0.50 giving the greatest spread.

**Discrimination index**  $D = \frac{(R_U - R_L)}{n}$ , where  $R_U$  and  $R_L$  are the numbers correct in the **upper 27%** and **lower 27%** groups, and n is the size of one group. Range -1 to +1. Ebel's guide:  $\geq 0.40$  **very good**; **0.30–0.39 good**; **0.20–0.29 marginal, revise**; **< 0.20 poor, discard**. A negative D means weak students did better than strong ones: check the key.

**Distractor analysis:** every distractor should attract some low scorers. A distractor nobody picks is dead and should be replaced.

## 5. The statistics you must be able to compute

**Mean, SD, normal-curve percentages and z-scores are the four tools behind every numeric question here.**

Scales of measurement (**Stevens, 1946**), mnemonic **NOIR**:

- **Nominal:** labels only (house, gender). Statistic: mode.
- **Ordinal:** ranks with unequal gaps (class rank). Statistic: median, Spearman's rho.
- **Interval:** equal units, arbitrary zero (test marks, °C). Mean, SD, Pearson's r.
- **Ratio:** true zero (height, time). Everything, including ratios.

**Central tendency:** mean (sum ÷ N; pulled by extreme scores), median (middle score; use with skewed or ordinal data), mode (most frequent; nominal data). For a moderately skewed distribution: **Mode = 3 Median – 2 Mean**.

**Dispersion:** range = highest – lowest; **quartile deviation**  $QD = (Q3 - Q1) \div 2$ ; mean deviation; **SD** =  $\sqrt{[\Sigma(X - M)^2 \div N]}$ ; **variance** =  $SD^2$ . SD is the most stable measure and goes with the mean; QD goes with the median.

**Normal probability curve:** bell-shaped, symmetrical, **mean = median = mode**, asymptotic to the base line, total area 1.00.

- $M \pm 1\sigma$ : **68.26%** of cases
- $M \pm 2\sigma$ : **95.44%**
- $M \pm 3\sigma$ : **99.73%**
- $M$  to  $+1\sigma$ : 34.13%;  $+1\sigma$  to  $+2\sigma$ : 13.59%;  $+2\sigma$  to  $+3\sigma$ : 2.14%; beyond  $+3\sigma$ : 0.13%.

**Skewness: positive** = tail to the right, most scores low,  $\text{mean} > \text{median} > \text{mode}$ : the test was **too hard**.

**Negative** = tail to the left, scores pile up high,  $\text{mean} < \text{median} < \text{mode}$ : the test was **too easy**. Pearson's  $Sk = 3(\text{Mean} - \text{Median}) \div \text{SD}$ . **Kurtosis:** leptokurtic (peaked), mesokurtic (normal), platykurtic (flat).

**Standard scores:**

- $z = (X - M) \div \text{SD}$ ; mean 0, SD 1; negative below the mean.
- $T = 50 + 10z$ ; mean 50, SD 10; no negatives.
- **Stanine:** 1 to 9, mean 5, SD 2.
- **Deviation IQ:** mean 100, SD 15 (Wechsler, and Stanford-Binet from the 5th edition); older Stanford-Binet editions used SD 16.  $\text{Ratio IQ} = \text{MA} \div \text{CA} \times 100$ .

**Percentile  $P_k$**  is the score below which  $k\%$  of the group falls; **percentile rank** is the percentage of the group below a given score. **P50 = median = Q2 = D5**;  $Q1 = P25$ ;  $Q3 = P75$ . Percentiles are ordinal, so they cannot be averaged.

**Correlation: Pearson's product-moment  $r$**  for interval data; **Spearman's rank-difference  $\rho$**  for ranks:  $\rho = 1 - 6\sum D^2 \div [N(N^2 - 1)]$ . Range  $-1$  to  $+1$ ; the sign gives direction, the size gives strength. Garrett's guide: 0.00–0.20 negligible; 0.20–0.40 low; 0.40–0.70 substantial; 0.70–1.00 high. Correlation never proves causation. Reliability and validity coefficients are both correlations.

## 6. Solved examples

**Example 1. Mean, median, mode.** Scores: 8, 10, 12, 12, 15, 17, 24. Mean =  $98 \div 7 = 14$ . Median:  $N = 7$ , position  $(7 + 1) \div 2 = 4$ th score = **12**. Mode = **12**. Mean > median, so the distribution is positively skewed; the 24 pulls the mean up. Shortcut: for odd  $N$  the median is the  $(N + 1)/2$ -th score of the sorted list.

**Example 2. Even  $N$  and QD.** Scores: 40, 55, 62, 70, 75, 90. Median =  $(62 + 70) \div 2 = 66$ . Mean =  $392 \div 6 = 65.3$ . If a distribution has  $Q1 = 48$  and  $Q3 = 72$ ,  $QD = (72 - 48) \div 2 = 12$ .

**Example 3. Empirical mode.** Mean 60, median 58. Mode =  $3(58) - 2(60) = 174 - 120 = 54$ . Mode < median < mean, so **positively skewed**: the test was hard for this group.

**Example 4. SD and variance.** Scores: 2, 4, 6, 8, 10.  $M = 6$ . Deviations:  $-4, -2, 0, 2, 4$ . Squares: 16, 4, 0, 4, 16, sum 40. Variance =  $40 \div 5 = 8$ . SD =  $\sqrt{8} = 2.83$ . Shortcut: adding a constant to every score leaves SD unchanged; multiplying by  $k$  multiplies SD by  $k$ . So 12, 14, 16, 18, 20 also has SD 2.83, and 4, 8, 12, 16, 20 has SD 5.66.

**Example 5. z and T scores.** Class mean 62, SD 8. Priya scores 74:  $z = (74 - 62) \div 8 = 1.5$ ;  $T = 50 + 10(1.5) = 65$ . Aman scores 54:  $z = -8 \div 8 = -1$ ;  $T = 40$ .

**Example 6. Comparing subjects.** Maths: M 40, SD 5, score 50. English: M 60, SD 10, score 75. Maths  $z = 10 \div 5 = 2.0$ . English  $z = 15 \div 10 = 1.5$ . The lower raw score in Maths is the better relative performance.

**Example 7. Normal-curve counts.** 1,000 students, M 60, SD 10. Between 50 and 70 ( $\pm 1\sigma$ ):  $68.26\% \approx 683$ . Between 60 and 70:  $34.13\% \approx 341$ . Above 80 ( $z = 2$ ):  $100 - 97.72 = 2.28\% \approx 23$ . Below 40 ( $z = -2$ ): also  $\approx 23$ .

**Example 8. Percentile rank from z.**  $z = +1 \rightarrow$  below it lie  $50 + 34.13 = 84\%$ , so  $PR \approx 84$ .  $z = -1 \rightarrow PR \approx 16$ .  $z = 0 \rightarrow PR 50$ .  $z = +2 \rightarrow PR \approx 98$ .

**Example 9. Spearman's rho.** Five students ranked in Maths 1, 2, 3, 4, 5 and in Science 2, 1, 4, 3, 5.  $D = -1, +1, -1, +1, 0$ ;  $\Sigma D^2 = 4$ .  $\rho = 1 - 6(4) \div [5(25 - 1)] = 1 - 24 \div 120 = 0.80$ , a high positive correlation.

**Example 10. Item analysis.** Try-out on 100 students; upper and lower 27% groups of 27 each. Item 7: 24 correct in the upper group, 9 in the lower.  $D = (24 - 9) \div 27 = 0.56$ , a very good item.  $p$  on the two groups =  $33 \div 54 = 0.61$ . If 60 of all 100 got it right,  $p = 0.60$ , inside the 0.30–0.70 band.

**Example 11. Spearman–Brown.** Split-half correlation  $r = 0.60$ . Full-test reliability =  $2(0.60) \div (1 + 0.60) = 1.20 \div 1.60 = 0.75$ . Tripling a test with  $r = 0.60$ :  $3(0.60) \div [1 + 2(0.60)] = 1.80 \div 2.20 = 0.82$ .

**Example 12. SEM and maximum validity.** SD 10, reliability 0.84.  $SEM = 10 \times \sqrt{1 - 0.84} = 10 \times \sqrt{0.16} = 10 \times 0.4 = 4$ . A score of 70 has a true score of **66–74** at 68% confidence and 62–78 at 95%. Maximum validity =  $\sqrt{0.84} \approx 0.92$ ; for  $r = 0.81$  it is exactly 0.90.

## 7. Assessment policy: RTE, NEP 2020, PARAKH

**CCE is a legal duty under RTE; NEP 2020 turns it into competency-based, 360-degree assessment steered by PARAKH.**

Timeline 2009 : RTE Act makes CCE mandatory  
 2019 : RTE Amendment, s.16 exams  
 2020 : NEP 2020, HPC, PARAKH  
 2023 : PARAKH set up under NCERT  
 2024 : PARAKH Rashtriya Sarvekshan held  
 2026 : CBSE two Class 10 exams

### RTE Act 2009

- **Section 29(2)(h):** comprehensive and continuous evaluation of the child's understanding and ability to apply knowledge. This is the legal root of CCE.
- **Section 30(1):** no board examination until elementary education is completed.
- **Section 16,** as substituted by the RTE (Amendment) Act 2019 (in force **1 March 2019**): a regular examination in **Class 5 and Class 8** at the end of every academic year; a child who fails gets additional instruction and a **re-examination within two months**; the appropriate Government may then allow the child to be held back; **no child may be expelled** before completing elementary education. The Central RTE (Amendment) Rules of

**December 2024** operationalised this for centrally run schools such as KVs and JNVs.

## **NEP 2020, "Transforming Assessment for Student Development"**

- **Para 4.34:** assessment shifts from summative and rote-based to regular, formative and competency-based.
- **Para 4.35:** the **Holistic Progress Card (HPC):** a **360-degree, multidimensional** report across cognitive, affective and psychomotor domains, combining **self-assessment, peer assessment and teacher assessment**. NCERT-PARAKH has published HPCs for the Foundational, Preparatory and Middle stages.
- **Paras 4.36–4.38:** Board exams for Grades 10 and 12 continue but are reformed to remove the coaching culture. They test core competencies and become lower-stakes: students may take them on **up to two occasions** in a school year, one main and one for improvement (para 4.38). CBSE did this for Class 10 in **2026:** the main exam in February–March, an optional second exam in May, and the **better score counts**.
- **Para 4.40:** school examinations in **Grades 3, 5 and 8**, conducted by the appropriate authority, testing basic learning outcomes.
- **Para 4.41:** **PARAKH** (Performance Assessment, Review, and Analysis of Knowledge for Holistic Development), a National Assessment Centre created as a **standard-setting body**. It was set up

under **NCERT in 2023**. Roles: norms and guidelines for assessment, guidance to boards and SCERTs, equivalence of boards, and the national survey.

**PARAKH Rashtriya Sarvekshan 2024** (the renamed National Achievement Survey)

- Held on **4 December 2024** in **Classes 3, 6 and 9**, the ends of the Foundational, Preparatory and Middle stages of the 5+3+3+4 design. About **21 lakh students**; results released **July 2025**. No newer round had been held as of October 2026.
- Earlier rounds: NAS 2017 (Classes 3, 5, 8) and NAS 2021, held 12 Nov 2021 (Classes 3, 5, 8, 10).
- Headline finding: average Maths performance falls from about **60% in Class 3 to about 37% in Class 9**.
- Other PARAKH products: the **School Quality Assessment and Assurance Framework (SQAAP)** and the **Equivalence of Boards** report.

Delhi context: DoE schools sit CBSE boards for Classes 10 and 12. For the middle classes, the DoE Examination Branch sets common question papers across government schools. **Mission Buniyaad** (since 2018) screens reading and numeracy and runs remedial classes: assessment for learning at system scale.

### Common traps

- **Create vs Evaluate**: in the 2001 revision Create is the top level; in 1956 Evaluation was. "Synthesis" belongs only to the original.

- **Reliable but invalid is possible; valid but unreliable is not.** Options that say the reverse are wrong.
- **Spearman-Brown corrects split-half**, not test-retest. KR formulas need right/wrong items; Cronbach's alpha handles rating scales.
- **Positive skew = hard test** (mean highest, tail to the right). Negative skew = easy test. Do not pair "positive" with "good".
- **68-95-99.7:**  $\pm 1\sigma$  is 68%, not 95%. Between the mean and  $+1\sigma$  is 34%, not 68%.
- **T-score has mean 50, SD 10;** z has mean 0, SD 1; stanine mean 5, SD 2; deviation IQ mean 100, SD 15.
- **Percentile vs percentage:** PR 90 means 90% of the group scored below you, not that you scored 90%.
- **Difficulty index:** a higher p means an easier item, not a harder one.
- **27%** is the standard upper/lower group size for D; options with 25% or 33% are distractors.
- **PRS 2024 classes are 3, 6, 9;** NAS 2017 was 3, 5, 8; NAS 2021 added 10.
- **RTE re-exam window is two months;** no-detention was ended by the **2019** Amendment, in force 1 March 2019, not by NEP 2020.
- **Section 29(2)(h)** gives CCE; **Section 30(1)** bars board exams at the elementary stage.
- **Scriven** coined formative/summative; **Glaser** coined norm/criterion-referenced; **Tyler** is the father of evaluation; **Stevens** gave NOIR.

## Remember for the exam

1. Evaluation = measurement + value judgement; Tyler linked objectives, experiences, evaluation.
2. Scriven 1967: formative/summative; Glaser 1963: norm-/criterion-referenced.
3. Bloom 1956 cognitive; Krathwohl 1964 affective (R-R-V-O-C); Simpson 1972 / Dave 1970 psychomotor.
4. Revised Bloom 2001: Remember, Understand, Apply, Analyse, Evaluate, Create; knowledge dimension adds Metacognitive.
5. Validity types: content, face, concurrent, predictive, construct; blueprint gives content validity.
6. Reliability methods: test-retest, parallel forms, split-half, KR-20/21, Cronbach's alpha, inter-rater.
7. Spearman-Brown:  $r_{full} = 2r \div (1 + r)$ .  $SEM = SD \times \sqrt{1 - r}$ . Max validity =  $\sqrt{r}$ .
8.  $p = \text{correct} \div \text{total}$  (target 0.30–0.70);  $D = (R_U - R_L) \div n$  with 27% groups;  $D \geq 0.40$  very good.
9. NOIR scales; Mode = 3 Median – 2 Mean;  $QD = (Q3 - Q1) \div 2$ ; variance =  $SD^2$ .
10. Normal curve: 68.26%, 95.44%, 99.73%; mean = median = mode.
11. Positive skew = hard test, mean > median > mode.
12.  $z = (X - M) \div SD$ ;  $T = 50 + 10z$ ; stanine mean 5 SD 2; deviation IQ SD 15.
13. Spearman  $\rho = 1 - 6\sum D^2 \div [N(N^2 - 1)]$ ;  $P50 = \text{median} = Q2$ .
14. RTE s.29(2)(h) CCE; s.30(1) no elementary board exam; s.16 (2019): Class 5 and 8 exams, re-exam within 2 months, no expulsion.

15. NEP 4.35 HPC (360-degree, self and peer assessment); 4.38 two board-exam occasions; 4.40 school exams in Grades 3, 5, 8; 4.41 PARAKH (NCERT, 2023); PRS 2024 on 4 Dec 2024, Classes 3, 6, 9, about 21 lakh students.

## Sources

- Right of Children to Free and Compulsory Education Act, 2009, as amended by the RTE (Amendment) Act, 2019; RTE (Amendment) Rules, 2024.
- National Education Policy 2020, paras 4.34–4.41 (Ministry of Education).
- NCERT–PARAKH, Holistic Progress Cards for the Foundational, Preparatory and Middle stages; PARAKH Rashtriya Sarvekshan 2024 National Report (July 2025).
- CBSE, Assessment and Evaluation Handbook; SAFAL handbook.
- IGNOU BES-127 *Assessment for Learning*, Blocks 1–4; NIOS D.El.Ed Course 502, Block 4 *Learning Assessment*.
- Bloom et al., *Taxonomy of Educational Objectives, Handbook I* (1956); Krathwohl, Bloom and Masia, *Handbook II* (1964); Anderson and Krathwohl, *A Taxonomy for Learning, Teaching, and Assessing* (2001).
- H.E. Garrett, *Statistics in Psychology and Education*; Thorndike and Hagen, *Measurement and Evaluation in Psychology and Education*.
- Current-affairs digest (research/05): CBSE Class 10 two-exam schedule, 2026.

## Quick facts

- Evaluation = measurement + value judgement; Ralph Tyler is the father of educational evaluation.
- Scriven (1967) coined formative/summative; Glaser (1963) coined norm-referenced/criterion-referenced.
- Bloom 1956 cognitive levels: Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation.
- Revised Bloom (Anderson & Krathwohl 2001): Remember, Understand, Apply, Analyse, Evaluate, Create; knowledge dimension adds Metacognitive.
- Affective domain: Krathwohl, Bloom, Masia 1964 (Receiving, Responding, Valuing, Organising, Characterising); psychomotor: Simpson 1972, Dave 1970, Harrow 1972.
- A reliable test may be invalid, but a valid test must be reliable; maximum validity = square root of reliability.
- Split-half reliability is corrected by Spearman-Brown:  $r_{full} = 2r / (1 + r)$ ;  $SEM = SD \times \sqrt{1 - r}$ .
- Item analysis: difficulty  $p = \text{correct} / \text{total}$  (target 0.30–0.70); discrimination  $D = (RU - RL) / n$  using upper and lower 27% groups;  $D \geq 0.40$  is very good.
- Normal curve: 68.26% within  $\pm 1$  SD, 95.44% within  $\pm 2$  SD, 99.73% within  $\pm 3$  SD; mean =

median = mode.

- Positive skew (mean > median > mode) means the test was too hard; negative skew means too easy.
- $z = (X - M) / SD$ ;  $T = 50 + 10z$ ; stanine mean 5, SD 2; deviation IQ mean 100, SD 15.
- Spearman rho =  $1 - 6\sum D^2 / [N(N^2 - 1)]$ ; Mode = 3 Median - 2 Mean; QD =  $(Q3 - Q1) / 2$ .
- RTE s.29(2)(h) mandates CCE; s.30(1) bars board exams in elementary classes; s.16 (2019 Amendment, in force 1 Mar 2019) allows Class 5 and 8 exams with re-exam within two months and no expulsion.
- NEP 2020 para 4.35: Holistic Progress Card (360-degree, self, peer and teacher assessment); para 4.38: board exams on up to two occasions; para 4.40: school exams in Grades 3, 5, 8; para 4.41: PARAKH as standard-setting body, set up under NCERT in 2023.
- PARAKH Rashtriya Sarvekshan 2024: held 4 Dec 2024 in Classes 3, 6 and 9, about 21 lakh students, results July 2025; CBSE held two Class 10 board exams for the first time in 2026.